Ch33： 8， 35， 39， 47， 59， 67

Basic score 4\` + problem 35 4'+ problem 39 2'

8. We note the points at which the curve is zero (θ_2 = 0° and 90°) in Fig. 33-31. We infer that sheet 2 is perpendicular to one of the other sheets at θ_2 = 0°, and that it is perpendicular to the *other* of the other sheets when θ_2 = 90°. Without loss of generality, we choose θ_1 = 0°, θ_3 = 90°. Now, when θ_2 = 35°, it will be $\Delta\theta$ = 35° relative to sheet 1 and $\Delta\theta'$ = 55° relative to sheet 3. Therefore,

$$\frac{I_f}{I_i} = \frac{1}{2}\cos^2(\Delta\theta)\cos^2(\Delta\theta') = 11\% .$$

35. (a) Since $c = \lambda f$, where λ is the wavelength and f is the frequency of the wave,

$$f = \frac{c}{\lambda} = \frac{3.0\times10^8}{5.0} = 0.6\times10^8\,\text{Hz} \qquad \text{0.5 point}$$

(b) The angular frequency is

$$\omega = 2\pi f = 2\pi(0.6\times10^8\text{ Hz}) = 3.8\times10^8\text{ rad/s.} \quad \text{0.5 point}$$

(c) The angular wave number is

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{5.0} = 1.3\text{ rad/m} \qquad \text{0.5 point}$$

(d) The magnetic field amplitude

$$B_m = \frac{E_m}{c} = \frac{215\text{V/m}}{3.0\times10^8\text{m/s}} = 7.2\times10^{-7}\text{T.} \quad \text{0.5 point}$$

(e) $\vec{B}$ must be in the positive z direction when $\vec{E}$ is in the positive y direction in order for $\vec{E}\times\vec{B}$ to be in the positive x direction (the direction of propagation). So the magnetic field oscillates parallel to z axis.

0.5 point

(f) The intensity of the wave is

$$I = \frac{E_m^2}{2\mu_0 c} = \frac{(215\text{V/m})^2}{2(4\pi\times10^{-7}\text{H/m})(3.0\times10^8\text{m/s})} = 61.3\text{ W/m}^2. \quad \text{0.5 point}$$

(g) Since the sheet is perfectly absorbing, the rate per unit area with which momentum is delivered to it is I/c, so

$$\frac{dp}{dt} = F = \frac{IA}{c} = \frac{(61.3\text{ W/m}^2)(2.0\text{ m}^2)}{3.0\times10^8} = 4.1\times10^{-7}\text{ N} = 0.41\ \mu\text{N} \quad \text{0.5 point}$$

(h) The radiation pressure is

$$p_r = \frac{I}{c} = \frac{(61.3\ \text{W/m}^2)}{3.0\times10^8} = 2.0\times10^{-7}\ \text{Pa} = 0.2\ \mu\text{Pa} \qquad \text{0.5 point}$$

39. When examining Fig. 33-61, it is important to note that the angle (measured from the central axis) for the light ray in air, θ, is not the angle for the ray in the glass core, which we denote θ_1 . The law of refraction leads to

$$n_1 \sin\theta_1 = n_{air} \sin\theta = \sin\theta \quad \text{0.5 point}$$

The angle of incidence for the light ray striking the coating is the complement of θ_1, which we denote as θ_{comp}, and recall that

$$\theta_{\text{comp}} = \frac{\pi}{2} - \theta_1 \qquad \text{0.5 point}$$

In the critical case, θ_{comp} must equal θ_c. Therefore,

$$\frac{n_2}{n_1} = \sin\theta_c = \sin\theta_{\text{comp}} = \cos\theta_1 = \sqrt{1-\sin^2\theta_1} = \sqrt{1-\left(\frac{1}{n_1}\sin\theta\right)^2} \qquad \text{0.5 point}$$

which leads to the result: $\sin\theta = \sqrt{n_1^2 - n_2^2}$. With n_1 = 1.58 and n_2 = 1.46, we obtain

$$\theta = \sin^{-1}\left(\sqrt{1.58^2 - 1.46^2}\right) = 37.2°. \quad \text{0.5 point}$$

47. The angle of incidence θ_B for which reflected light is fully polarized is given by Eq. 33-48 of the text. If n_1 is the index of refraction for the medium of incidence and n_2 is the index of refraction for the second medium, then

$$\theta_B = \tan^{-1}(n_2/n_1) = \tan^{-1}(1.71/1.33) = 52.1°.$$

59. (a) The average rate of energy flow per unit area, or intensity, is related to the electric field amplitude E_m by $I = E_m^2 / 2\mu_0 c$, so

$$E_m = \sqrt{2\mu_0 cI} = \sqrt{2\left(4\pi\times10^{-7}\ \text{H/m}\right)\left(2.998\times10^8\ \text{m/s}\right)\left(28\times10^{-6}\ \text{W/m}^2\right)}$$
$$= 0.145\ \text{V/m}.$$

(b) The amplitude of the magnetic field is given by

$$B_m = \frac{E_m}{c} = \frac{0.145\,\text{V/m}}{2.998\times10^8\,\text{m/s}} = 4.8\times10^{-10}\,\text{T}.$$

(c) At a distance r from the transmitter, the intensity is $I = P/2\pi r^2$, where P is the power of the transmitter over the hemisphere having a surface area $2\pi r^2$. Thus

$$P = 2\pi r^2 I = 2\pi\left(10\times10^3\,\text{m}\right)^2\left(28\times10^{-6}\,\text{W/m}^2\right) = 1.76\times10^4\,\text{W}.$$

67. (a) Since the incident light is unpolarized, half the intensity is transmitted and half is absorbed. Thus the transmitted intensity is I = 3.25 mW/m^2. The intensity and the electric field amplitude are related by $I = E_m^2/2\mu_0 c$, so

$$E_m = \sqrt{2\mu_0 c I} = \sqrt{2(4\pi\times10^{-7})(3.0\times10^8)(3.25\times10^{-3})} = 1.6\ \text{V/m}$$

(b) The radiation pressure is $p_r = I_a/c$, where I_a is the absorbed intensity. Thus

$$p_r = \frac{I}{c} = \frac{3.25\times10^{-3}}{3.0\times10^8} = 1.1\times10^{-11}\ \text{Pa}$$